INGE LEHMANN

1888–1993

Seismologist Inge Lehmann's precise data analysis revealed a secret at the heart of the Earth—its solid core.

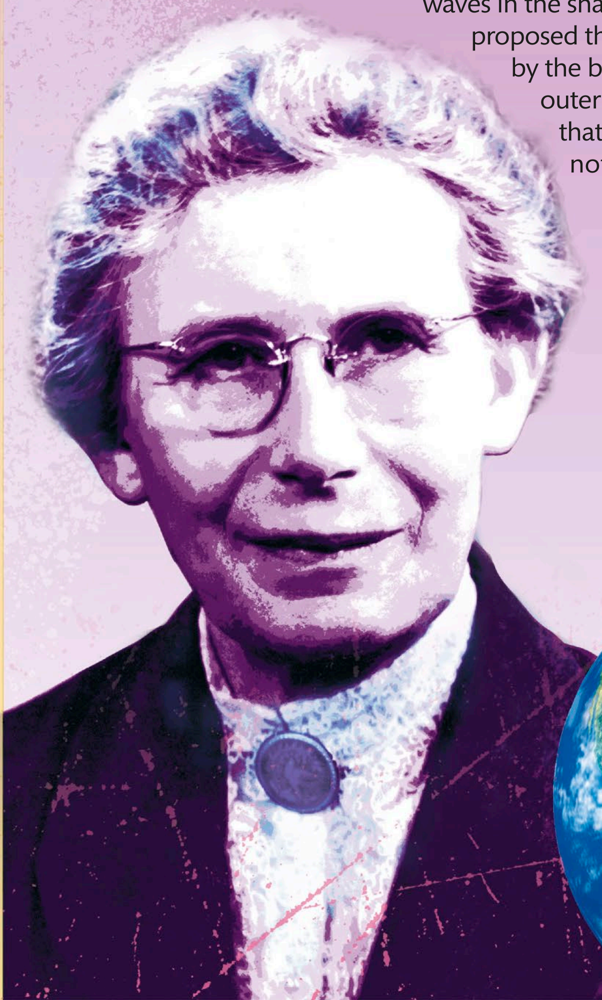
Born in Copenhagen, Denmark, Inge Lehmann gained a degree in mathematics and went on to study earthquakes. Scientists knew that shock waves from an earthquake—P, or primary, waves and S, or secondary, waves—traveled through the Earth and could be detected on the other side of the planet. They had also observed shadow zones, where few waves were detected, and thought they occurred because the shock waves were deflected by the Earth's core, which was liquid. However, this did not explain the occurrence of weak P waves in the shadow zone. In 1936, Lehmann proposed that these P waves were deflected by the boundary between the inner and outer regions of the Earth's core and that the inner region was solid and not liquid as previously thought.

MILESTONES

SEISMIC SHOCKS
Analyzes the data from a huge earthquake in New Zealand in 1929, and interprets the P waves.

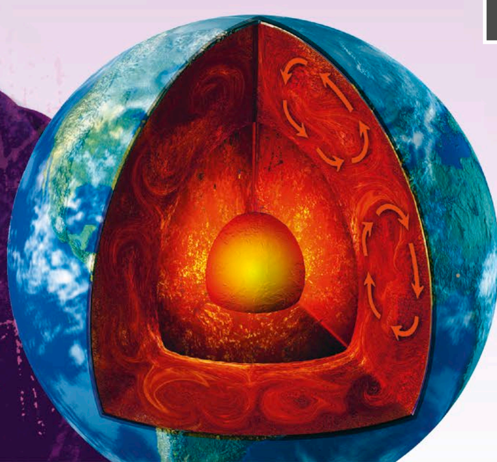
BREAKTHROUGH PAPER
Publishes her "P" paper in 1936, demonstrating that the Earth must have an inner core.

CORE BOUNDARY
In 1959, discovers another boundary, this one 137 miles (220 km) below the Earth's surface.



"The **master** of a **black art.**"

Bertha Swirles, 1994



Lehmann discovered *that the core of the Earth is more intriguing than was thought because the inner core rotates separately.*